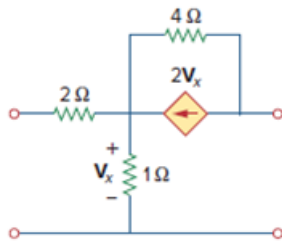


Problem

Repeat Prob. 19.26 using *PSpice* or *MultiSim*.

Calculate $[y]$ for the two-port in Fig. 19.85.

Figure 19.85



Step-by-step solution

Step 1 of 10

Draw the circuit diagram as shown below.

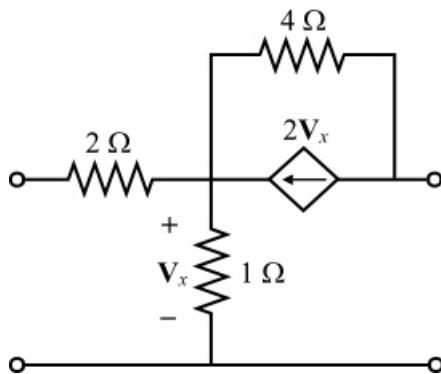


Figure 1

Step 2 of 10

Re-draw Figure 1, with node parameters.

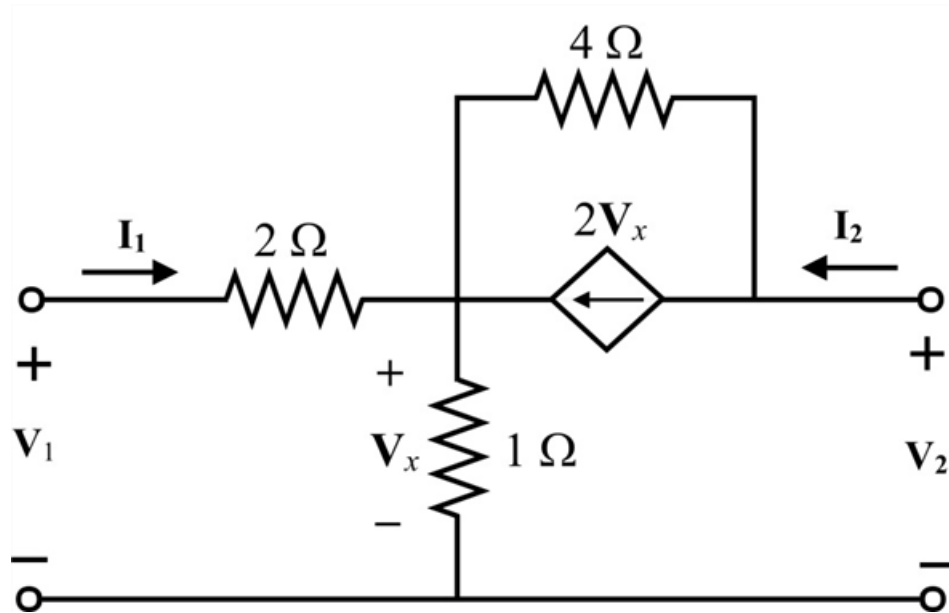


Figure 2

Step 3 of 10

Write the y-parameters.

$$y_{11} = \left. \frac{I_1}{V_1} \right|_{V_2=0} \quad y_{12} = \left. \frac{I_1}{V_2} \right|_{V_1=0}$$

$$y_{21} = \left. \frac{I_2}{V_1} \right|_{V_2=0} \quad y_{22} = \left. \frac{I_2}{V_2} \right|_{V_1=0}$$

Step 4 of 10

Calculate y_{11} and y_{21} .

(i) Set $V_2 = 0$, $V_1 = 1$ V.

Draw Pspice to obtain y_{11} and y_{21} .

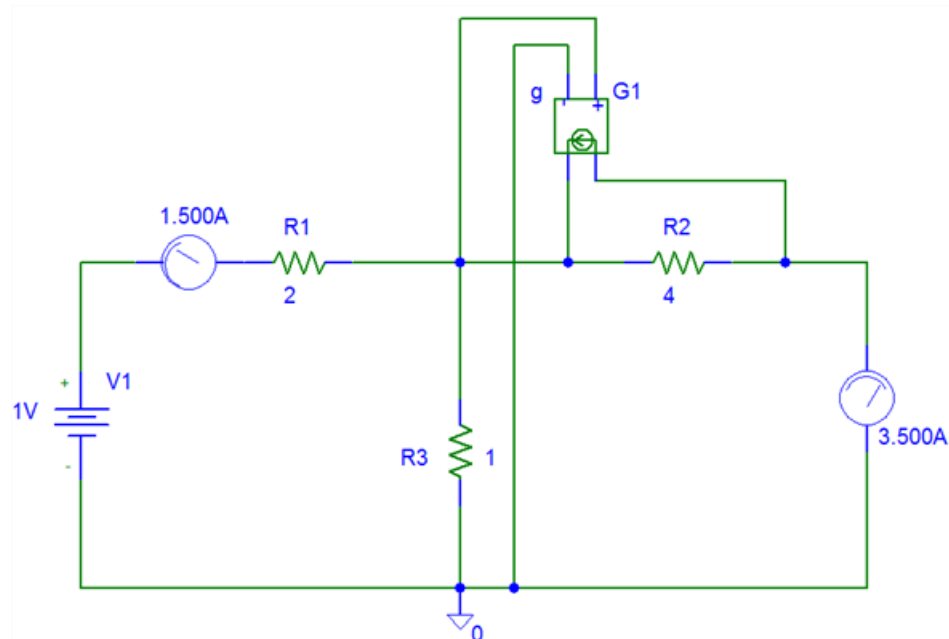


Figure 3

From Figure 3, the following Pspice results are obtained.

VOLTAGE SOURCE CURRENTS

NAME CURRENT

V_V1 -1.500E+00

v_V3 3.500E+00

v_V2 1.500E+00

From the results of Pspice, Calculate y_{11} and y_{21} values.

Calculate y_{11} value.

$$\begin{aligned} y_{11} &= \frac{I_1}{V_1} \\ &= \frac{1.5}{1} \\ &= 1.5 \text{ S} \end{aligned}$$

Therefore, the value of y_{11} is 1.5 S.

Calculate y_{21} value.

$$\begin{aligned} y_{21} &= \frac{I_2}{V_1} \\ &= \frac{3.5}{1} \\ &= 3.5 \text{ S} \end{aligned}$$

Therefore, the value of y_{21} is 3.5 S.

Calculate y_{12} and y_{22} .

(ii) Set $V_2 = 1 \text{ V}$, $V_1 = 0$.

Draw Pspice to obtain y_{12} and y_{22} .

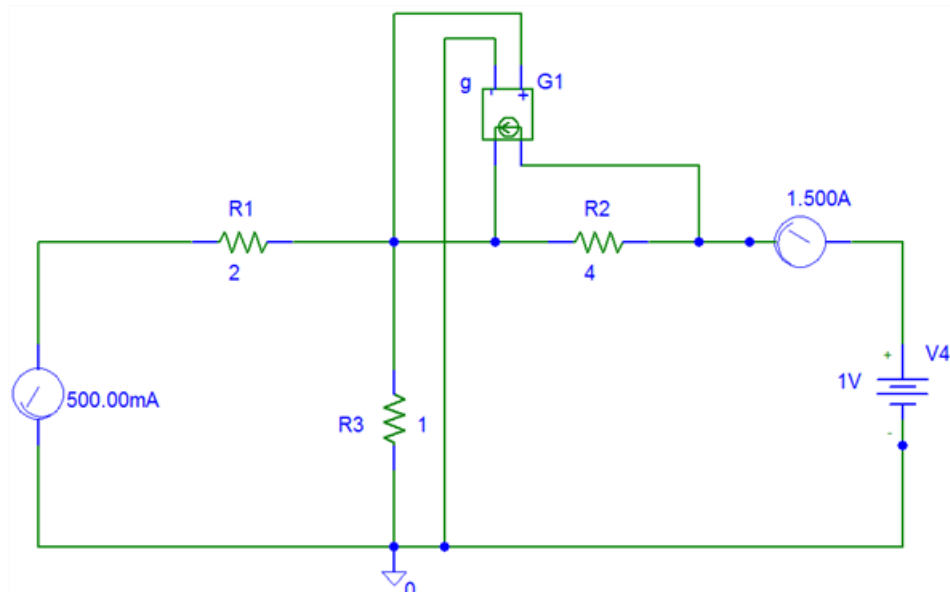


Figure 4

From Figure 4, the following Pspice results are obtained.

VOLTAGE SOURCE CURRENTS NAME CURRENT

Step 8 of 10

V_V4 1.500E+00 v_V3 1.500E+00 v_V2 -5.000E-01

Step 9 of 10

From the results of Pspice, Calculate y_{12} and y_{22} values.

Calculate y_{12} value.

$$\begin{aligned} y_{12} &= \frac{I_1}{V_2} \\ &= \frac{-0.5}{1} \\ &= -0.5 \text{ S} \end{aligned}$$

Therefore, the value of y_{12} is -0.5 S .

Calculate y_{22} value.

$$\begin{aligned} y_{22} &= \frac{I_2}{V_2} \\ &= \frac{1.5}{1} \\ &= 1.5 \text{ S} \end{aligned}$$

Therefore, the value of y_{22} is 1.5 S .

Step 10 of 10

Therefore, the y -parameters are:

$$[y] = \begin{bmatrix} 1.5 & -0.5 \\ 3.5 & 1.5 \end{bmatrix} \text{ S}$$